

Fig. 3: Actual-size PCB layout of the electronic security system

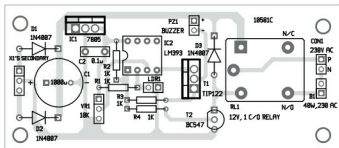


Fig. 4: Component layout of the PCB

Signal output from LDR1 is fed to IC2. Output from IC2 activates the alarm when light falls on LDR1 during a burglary attempt. At the

same time, bulb B1 glows as a visual alert to thwart the burglary attempt.

Case 1. If reference voltage at pin 3 of IC2 is less than sensor voltage at pin 2 of IC2, output goes low automatically. This indicates that no light is falling on LDR1. Low output of IC2 is fed to T1 and T2. As a result, both are in cut-off state and so

no current flows through the buzzer, keeping it in silence mode. Similarly, no current flows in the relay, and so B1 does not glow.

Case 2. If reference voltage at pin 3 of IC2 is greater than sensor voltage at pin 2 of IC2, output goes high automatically. This indicates that light is falling on LDR1. High output of IC2 is fed to T1 and T2. As a result, both transistors are in on (active) state. Current flows through the buzzer and relay. The buzzer produces a loud alarm to alert you or people around you and, at the same time, B1 glows as the circuit completes through relay RL1's contacts.

Construction and testing

An actual-size, single-side PCB for the electronic security system is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable plastic box. **EFY**



Pamarthi Kanakaraja is assistant professor in Usha Rama College of Engineering and Technology, Andhra Pradesh

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